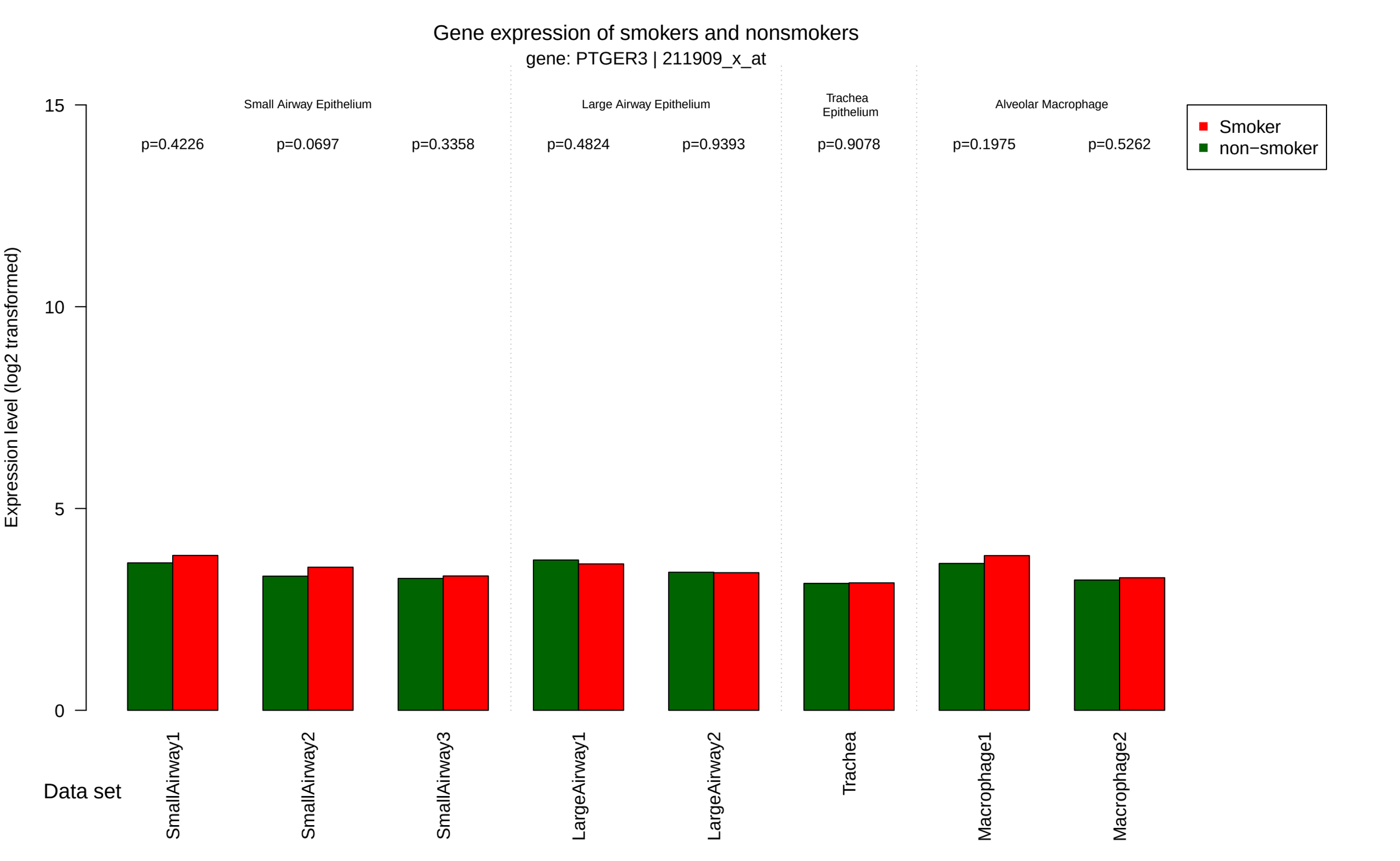
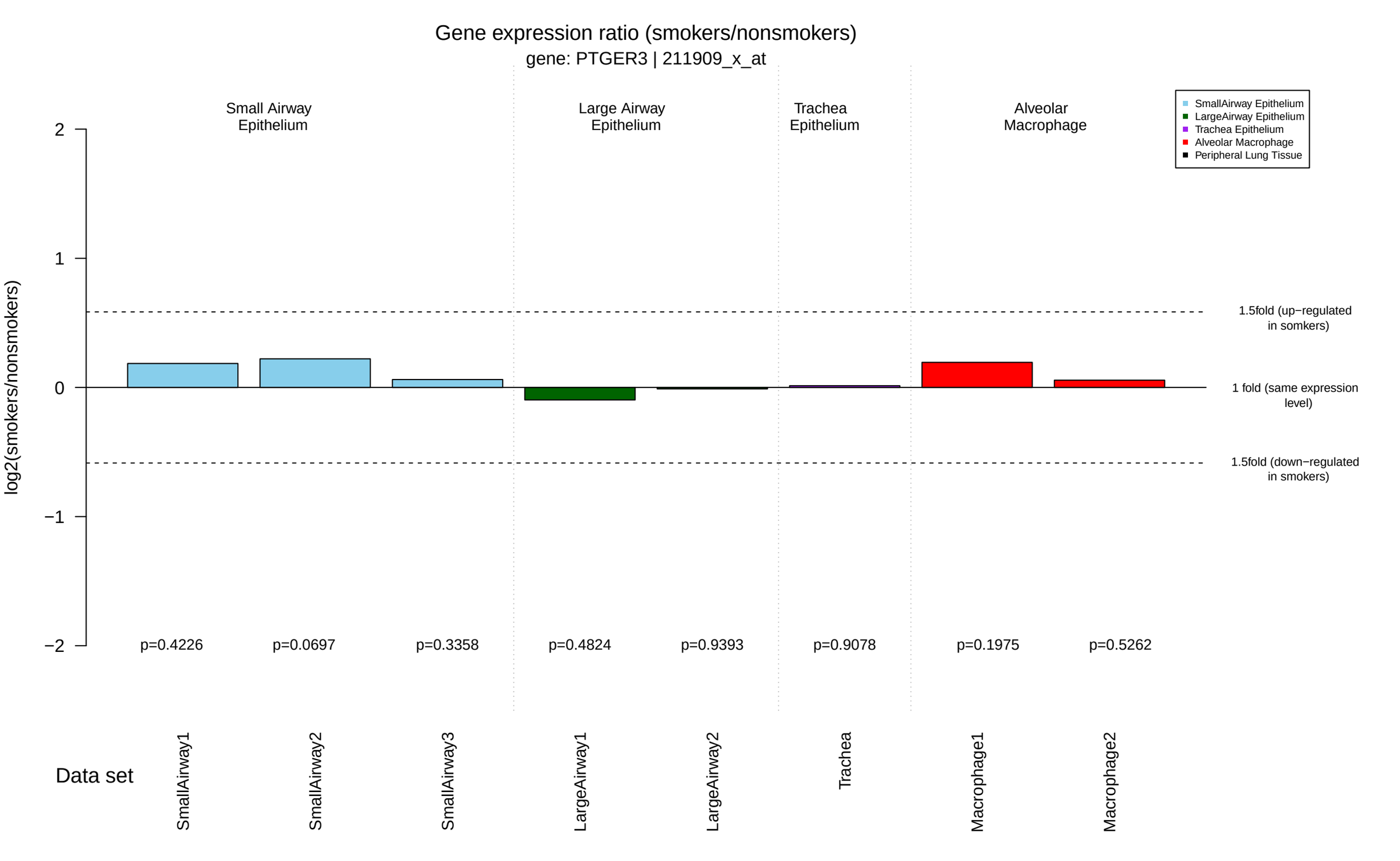
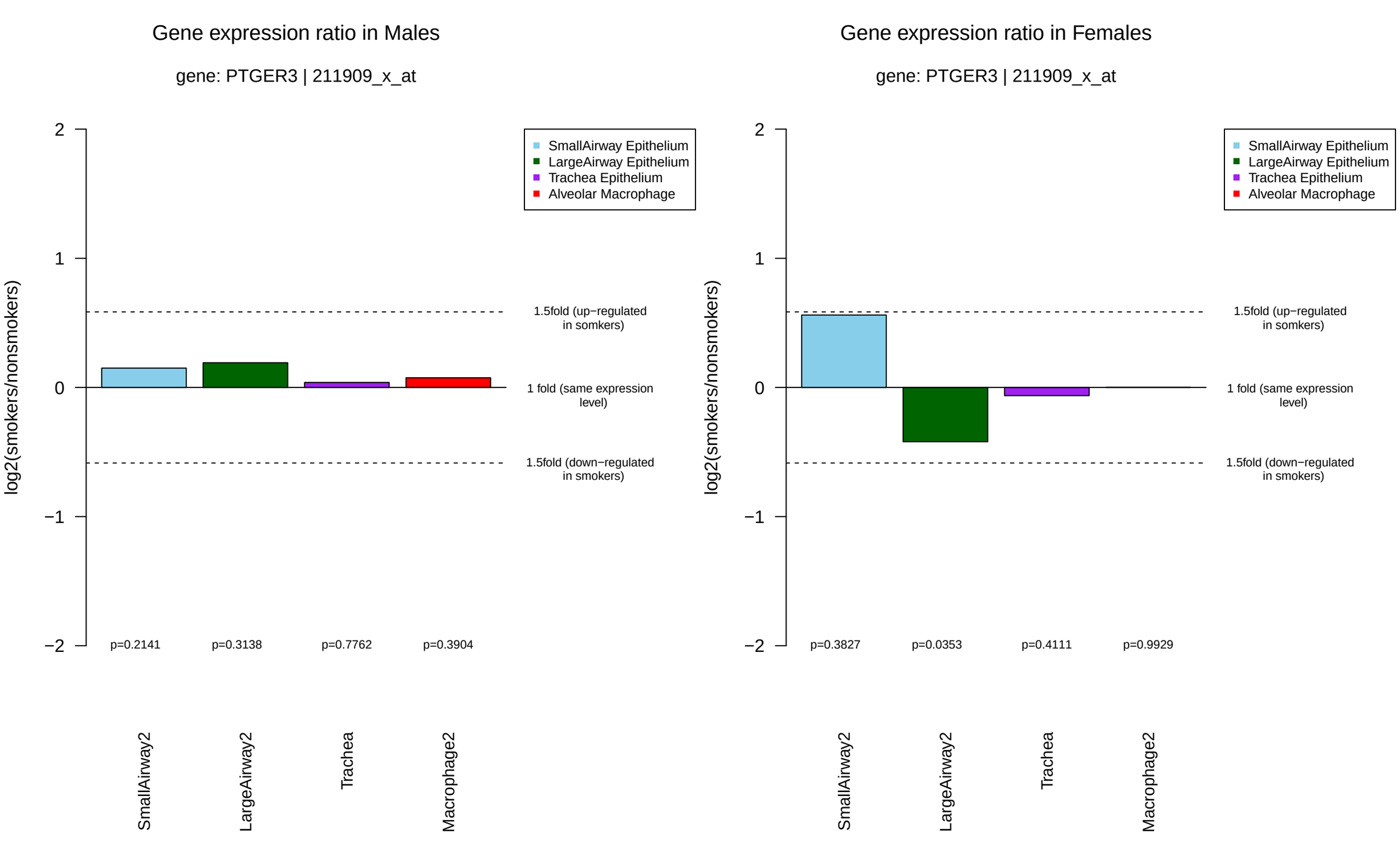
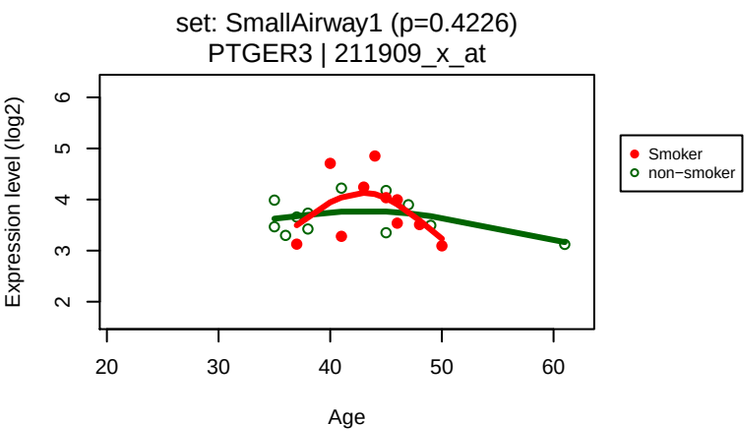
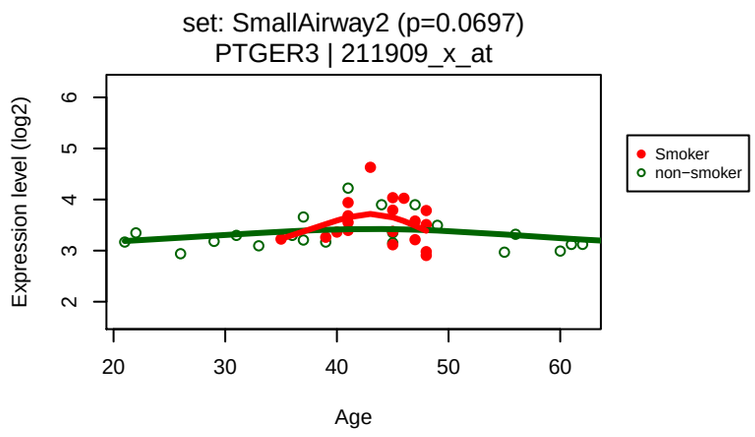
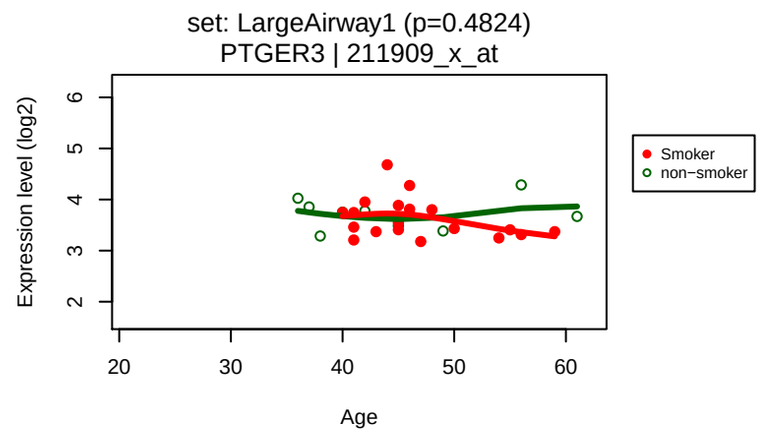

Gene expression of smokers and nonsmokers
gene: PTGER3 | 211909_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4226
p=0.0697
p=0.3358
p=0.4824
p=0.9393
p=0.9078
p=0.1975
p=0.5262
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

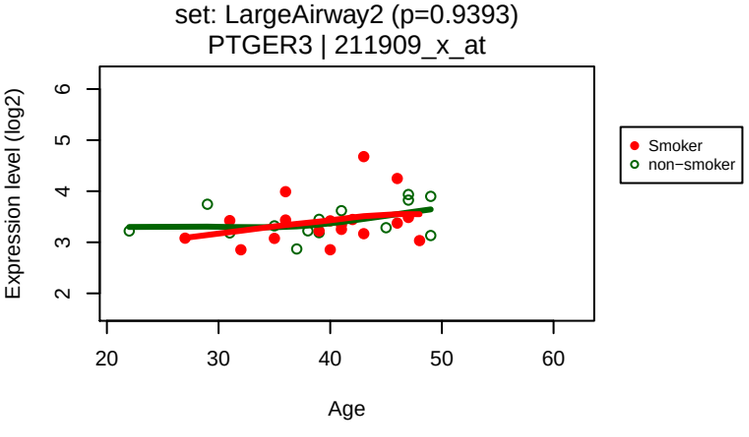

Gene expression ratio (smokers/nonsmokers)
gene: PTGER3 | 211909_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4226
p=0.0697
p=0.3358
p=0.4824
p=0.9393
p=0.9078
p=0.1975
p=0.5262
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

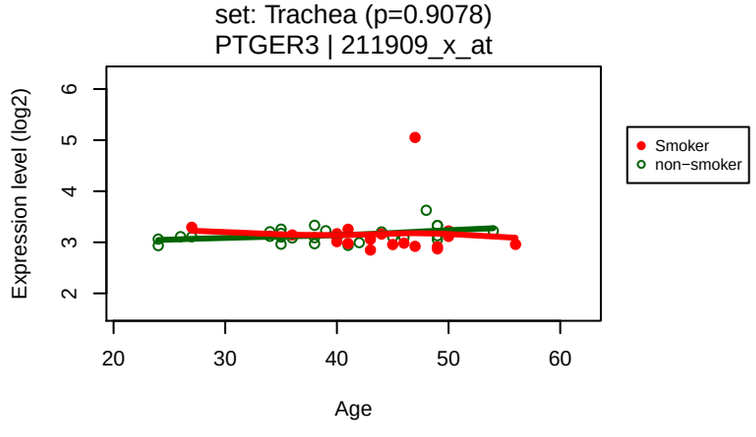

Gene expression ratio in Males
gene: PTGER3 | 211909_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2141
p=0.3138
p=0.7762
p=0.3904
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTGER3 | 211909_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3827
p=0.0353
p=0.4111
p=0.9929
SmallAirway2
LargeAirway2
Trachea
Macrophage2

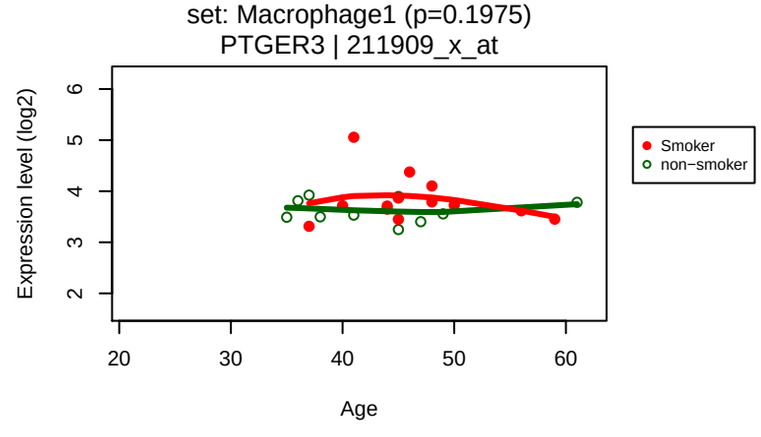

set: SmallAirway1 (p=0.4226)
PTGER3 | 211909_x_at
Expression level (log2)
Age
Smoker
non−smoker

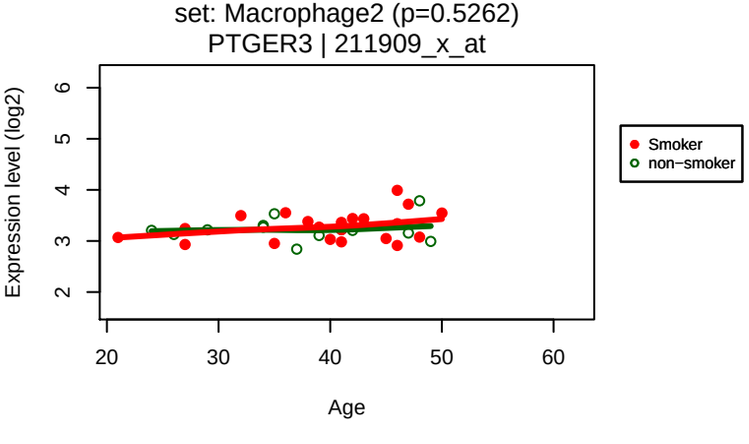

set: SmallAirway2 (p=0.0697)
PTGER3 | 211909_x_at
Expression level (log2)
Age
Smoker
non−smoker

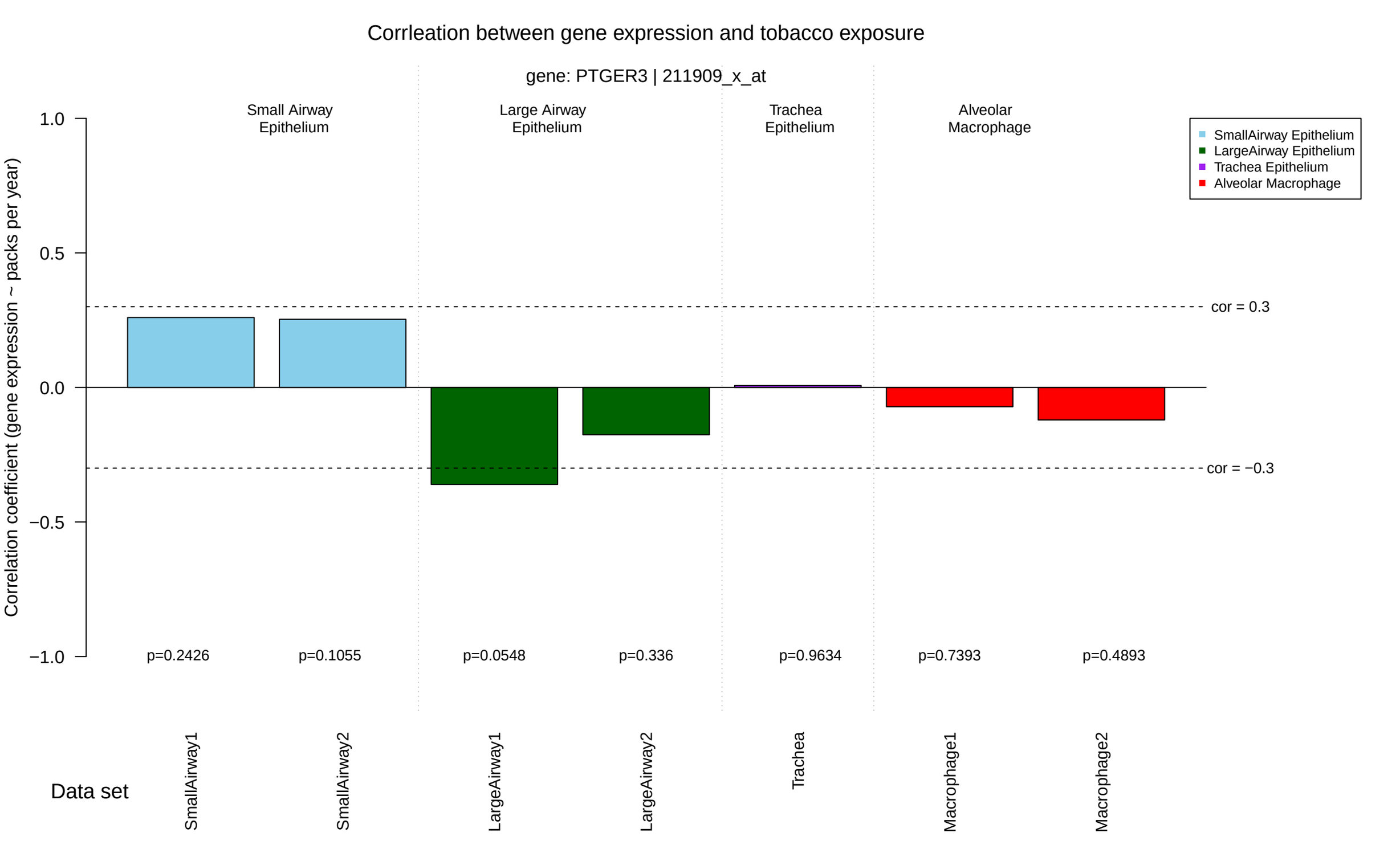

set: LargeAirway1 (p=0.4824)
PTGER3 | 211909_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9393)
PTGER3 | 211909_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9078)
PTGER3 | 211909_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1975)
PTGER3 | 211909_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5262)
PTGER3 | 211909_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTGER3 | 211909_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2426
p=0.1055
p=0.0548
p=0.336
p=0.9634
p=0.7393
p=0.4893
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2